

Digilians

Data Engineering

Data Engineering

- Data engineering is the process of designing and managing systems to collect, store, and analyze large amounts of data.
- It involves creating data pipelines to move data between systems and ensuring data is accurate and accessible.
- Data engineers use various tools and technologies to support data-driven decision-making.

Why to Learn Data Engineering?

- Learning data engineering helps you build and manage systems that handle large amounts of data efficiently.
- It opens up job opportunities in tech and businesses that rely on data.
- With these skills, you can support data-driven decisions and improve company operations.

Data Engineering Features

- Data engineering features include creating data pipelines to move and process data.
- It ensures data is clean, accurate, and available for analysis.
- It also involves using tools and technologies to handle large-scale data efficiently.

Who Should Learn Data Engineering?

- People who enjoy working with data and technology should learn data engineering.
- It is great for those wanting to build and manage data systems.
- This field is ideal for anyone aiming for a career in tech, analytics, or data science.

Prerequisites to Learn Data Engineering

- To learn data engineering, you should know basic programming, especially Python or Java.
- Understanding databases and SQL is also important.
- Basic knowledge of data processing and storage concepts helps too.

Data Engineering Jobs and Opportunities

- Data engineering offers jobs like data engineer, data architect, and ETL developer.
- These roles are in high demand in tech companies, finance, and healthcare.
- With data engineering skills, you can work on exciting projects and earn a good salary.

What is data engineering?

- Data engineering is the process of designing, building, and maintaining systems for collecting, storing, and processing large volumes of data.
- It involves creating data pipelines and ensuring data is clean, accurate, and accessible.

Why is data engineering important?

- Data engineering is important because it ensures that data is available and reliable for analysis and decision-making.
- It helps organizations make informed decisions, optimize operations, and gain insights from their data.

What skills do I need to become a data engineer?

To become a data engineer, you need strong programming skills in languages like Python or Java.

Knowledge of databases and SQL is essential, along with familiarity with data processing tools like Hadoop and Spark.

What tools do data engineers use?

- Data engineers use various tools to manage and process data, including Hadoop for distributed storage and processing, Spark for big data analytics, SQL for database management, and ETL tools for data integration.

How does data engineering differ from data science?

- Data engineering focuses on building and maintaining the infrastructure for data collection, storage, and processing.
- In contrast, data science involves analyzing and interpreting data to derive insights and build predictive models.

Can I learn data engineering without a degree?

- Yes, you can learn data engineering through online courses, coding bootcamps, and self-study.
- Practical experience through projects and internships can also help you gain the necessary skills without a formal degree.

What is a data pipeline?

- A data pipeline is a series of processes that move data from one system to another, often involving steps like data extraction, transformation, and loading (ETL).
- It ensures data flows smoothly and is ready for analysis.

What industries need data engineers?

- Data engineers are needed in various industries, including technology, finance, healthcare, retail, and manufacturing.
- Any industry that relies on data for decision-making and operations can benefit from data engineering.

How much do data engineers earn?

- Data engineers typically earn high salaries, ranging from \$80,000 to \$150,000 per year, depending on their experience, location, and the company's size.
- Experienced engineers in major tech hubs can earn even more.

What is the future of data engineering?

- The future of data engineering is promising, with increasing demand as companies continue to adopt data-driven strategies.
- Emerging technologies like artificial intelligence and machine learning will further enhance the field's growth.

- In Data Engineering, data processing and analysis are carried out with the help of high-performance computing.
- In the context of computing, the data engineering approach adopted is called data flow programming.
- It represents computations as a directed graph, in which nodes are different operations, and edges represent the flow of data.
- Implementations like increment computing determines the efficiency of data processing.

- Data Engineering involves creating systems and tools to manage the access and flow of the information.
- Here, the data is easily accessible, well-maintained and ready for analysis.
- Data engineers build and manage the data infrastructure, making it easier for data analyst and scientists to work with the data.

- Data Engineering involves developing, implementing and maintaining systems that transform unique data into high-quality reliable information.
- This information is used for various purposes, such as machine learning and analysis.
- Data engineering combines aspects of security, data management, DataOps and data architecture and software engineering.
- A data engineering is a system that makes the source available for use in analysis or machine learning.

Data Engineering

- Data is stored in a wide range of application and one key factor in deciding how to store data is its purpose.
- Data Engineers optimize storage by compressing, archiving and partitioning data.

- Structured data often requires online transaction processing, for which databases are typically used.
- Relational databases with strong ACID guarantees and SQL queries were once common.
- NoSQL databases have gained popularity due to their ability to scale horizontally, even though they sacrifice ACID guarantees and reduce the object- relational mismatch.

- When structured data needs analytical processing instead of transaction processing, data warehouses are typically used.
- They support large scale data analysis and manage data flow from databases.
- Data engineers, business analysts and data scientists access data warehouses using tools like SQL or business intelligence software.

Data Engineering Tools

- A data lake is a centralized repository that allows massive amounts of data, whether structured from relational databases or semi-structured and unstructured data or binary data to be stored.
- This could be done through services available in the public cloud like Microsoft, Amazon or Google.
- If the data is less specified, then they are stored as files.

There are different options such as:

- Object storage manages data using the help of metadata, sometimes assigning an unique key to every file, such as Universally Unique identifier(UUID).
- Block storage divides data into equally sized chunks, which often correspond to hard drives or solid-state drives.
- File systems organize data hierarchically using nested folders.

- Big data is very much popular and captures the interest of many companies.
- Often, companies use big tools for small data problems, deploying complex systems for minimum data.
- This trend is driven by various marketing strategies for big data tools.

- Data Scientists usually create production data systems, but they often work inefficiently due to limited support and resources from data engineers.
- Data scientists should focus their time on analytics, machine learning and experimentation.
- When data engineers handle the foundation tasks, they create a solid base that allows data scientists to excel in their roles.

THANK
you